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Problem Set 5

I. THE BLACK–SCHOLES PRICE AS A FUNCTION OF S_t AND DELTA

This question should help you to understand the Black–Scholes delta better.

1. Fix $r = 5\%$, $K = 50$, $\sigma = 30\%$, and $T - t = 1$. (You should use "absolute references" to these cells in Excel.) Create a column with values of $S_t = 1, 2, ..., 100$.
2. Populate a column with the Black–Scholes price for each value of S_t. (Remember to use the function `NORM.DIST()`.)
3. Make a graph of the call price as a function of S_t. To do this you can use Excel's line chart.
4. Create a column with the Black–Scholes delta, $\Delta = \Phi(d_1)$ for each value of S_t.
5. Use the approximation to the derivative

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

   to calculate a numerical approximation of delta directly from the call price (i.e. do not use calculus or the formula). Use $h = 1$ and calculate the approximation for $S_t = 1, 2, ..., 99$.
6. Calculate the difference between your answers in questions 4 and 5 for $S_t = 1, 2, ..., 99$.

II. Pricing by Monte Carlo Simulation

This question should introduce you to Monte Carlo methods and enrich your understanding of the FTAP. Throughout, use $r = 5\%$, $K = 50$, $\sigma = 30\%$, $T - t = 1$, and $S_t = 50$.

1. The Excel function `RAND()` generates a random number from the uniform distribution on [0,1]. Populate a column with 1000 random numbers from this distribution.

2. Make a histogram of your draws from the uniform distribution and confirm it has approximately the square shape of the uniform density.[1]

3. The Excel function `NORMSINV()` computes the inverse of the standard normal distribution function. Populate a column with this function applied to the uniform output. You should have a column with 1000 draws of `NORMSINV(RAND())`.

4. Make a histogram of your draws from `NORMSINV`. Show that it has approximately the bell shaped curve of the normal density.[2]

5. For each value of your draws above, create a column with the associated stock price,

$$S_T = S_t e^{\left(r - \frac{1}{2}\sigma^2\right)(T-t) + \sigma\sqrt{T-t}\,z}$$

   where $z \sim \mathrm{N}[0,1]$.

6. Create a histogram of the stock price.

7. For each of the values of the stock price, create a column with the associated call payoffs.

8. Calculate the discounted average of the call payoffs to approximate $e^{-r(T-t)}\mathbb{E}_t^{\mathbb{Q}}\left[(S_T - K)^+\right]$.

9. Use the FTAP and your answer to approximate the call price at date t. Compare your answer to the output of the Black–Scholes formula. Are the numbers close? Why?

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[1] To make a histogram, you can proceed as follows. After you have your column of a 1000 random numbers, add another column with bin sizes, e.g., $0.1, 0.2, ..., 0.9, 1$. Then go to Data Analysis under the Data tab and select the histogram tool. In the Input Range field, put the cells with random numbers. In the Bin Range field, put the cells with your bin sizes. Check the Chart Output box at the bottom of the window. Then click OK.

[2] To see why formally, suppose that $\mathbf{U}$ is distributed uniformly on [0,1], written $\mathbf{U} \sim \mathrm{U}[0,1]$. In other words, if $u \in [0,1]$, then $\mathbb{P}\{\mathbf{U} \leq u\} = u$. Note that since $\Phi(z) \in (0,1)$, we can say that

$$\mathbb{P}\{\mathbf{U} \leq \Phi(z)\} = \Phi(z).$$

We can write $\mathbb{P}\{\mathbf{U} \leq \Phi(z)\} = \mathbb{P}\left\{\Phi^{-1}(\mathbf{U}) \leq z\right\}$. Combined with the above that says that $\Phi^{-1}(\mathbf{U})$ is normal.

III. Comparison with Data

Here we compare the output of the Black–Scholes model with the market prices, using an S&P 500 index tracker (SPY), options on SPY, and the VIX. You can get data, e.g., from Yahoo Finance:

https://finance.yahoo.com/quote/SPY?p=SPY

https://finance.yahoo.com/quote/SPY/options?p=SPY

https://finance.yahoo.com/quote/%5EVIX/

1. What is the current share price of SPY?
2. Now consider two puts. Choose the closest maturity to one month and one strike close to at the money and one deep out of the money. What are the prices of each? Use the midpoint of the bid and ask quotes and record the observation time, expiration date, and strikes.
3. What is the current value of the VIX?
4. If you use $\sigma = \text{VIX}/100$ in the Black–Scholes formula, what are the predicted prices of the put options you chose in question 2?

   NB: A quoted VIX value of 20 corresponds to $\sigma = 0.20$. Measure $T - t$ in years. SPY pays dividends and its options are American, whereas the formula used here assumes European options without dividends. VIX is based on S&P 500 index options and is only an approximate volatility input for these SPY options. These differences can contribute to discrepancies between model and market prices.

   NB: The formula depends on the risk-free rate r. You can use the yield data from the Treasury:

   https://www.treasury.gov/resource-center/data-chart-center/interest-rates/Pages/TextView.aspx?da

   However, the value of r you choose should not affect your answer much (why?).
5. Often, the Black–Scholes formula gives a good approximation of near-the-money options, but underestimates the price of out-of-the-money puts. Why might that be? Is that consistent with your analysis above?